

Problem

The voltage drop across a 1.5-kW toaster that draws 12 A of current is:

- (a) 18 kV
- (b) 125 V
- (c) 120 V
- (d) 10.42 V

Step-by-step solution

Step 1 of 2

Power absorbed or supplied by an element is the product of the voltage across the element and the current through it.

$$p = vi$$

Step 2 of 2

Substitute 1.5 kW for p and 12 for i in the equation.

$$1.5 \text{ kW} = v(12)$$

Solve for voltage v .

$$\begin{aligned} v &= \frac{1.5 \text{ kW}}{12} \\ &= 125 \text{ V} \end{aligned}$$

Thus, correct option is **b**.